

TOPIC 8: SIMPLE EQUATIONS INVOLVING FRACTIONS

1. Method

To solve equations with fractions: multiply through by the LCM of the denominators to clear fractions; expand and simplify; collect like terms; solve for the unknown; check by substitution.

2. Example

Solve: $(x/2) + (x/3) = 5$. LCM of 2 and 3 is 6. Multiply by 6: $3x + 2x = 30 \rightarrow 5x = 30 \rightarrow x = 6$. Check: $6/2 + 6/3 = 3 + 2 = 5$.

3. Word Problems

Define the unknown; write the equation from the problem statement (often involving parts or rates); clear fractions and solve; interpret the answer in words.

4. Summary

Clear denominators using the LCM, then solve as a normal linear equation. Always check your solution in the original equation.

End of Topic 8